Chapter 1: Limits, Continuity, and the Foundations of Calculus

Course: Advanced Mathematics & Calculus Foundations | **Target Level**: High School / Undergraduate

1.1 Intuitive Concept of a Limit

In calculus, the limit is the foundational concept upon which differentiation and integration rest. Intuitively, when we say the limit of $f(x)$ as x approaches c equals L, written as $\lim_{x \to c} f(x) = L$, we mean that we can make the value of $f(x)$ arbitrarily close to L by taking x sufficiently close to c, but not equal to c.

Consider the function $f(x) = (x^2 - 1) / (x - 1)$. At $x = 1$, $f(1)$ is undefined ($0/0$ indeterminate form). However, for $x \neq 1$, factoring gives $f(x) = (x-1)(x+1)/(x-1) = x+1$. As x approaches 1 from either side, $f(x)$ approaches 2. Thus, $\lim_{x \to 1} (x^2 - 1)/(x - 1) = 2$.

1.2 The Formal Epsilon-Delta Definition

The rigorous Cauchy-Weierstrass definition states that $\lim_{x \to c} f(x) = L$ if and only if:

```
For every epsilon > 0, there exists a delta > 0 such that whenever 0 < |x - c| <
delta, it follows that |f(x) - L| < epsilon.
```

Here, epsilon represents the maximum allowable error bound in the output space, while delta represents the corresponding required proximity in the input domain.

1.3 One-Sided Limits and Continuity

A two-sided limit exists if and only if the left-hand limit and right-hand limit both exist and are strictly equal:

```
lim_{x->c} f(x) = L <==> lim_{x->c^-} f(x) = L and lim_{x->c^+} f(x) = L
```

A function $f(x)$ is continuous at $x = c$ if: (1) $f(c)$ is defined, (2) $\lim_{x \to c} f(x)$ exists, and (3) $\lim_{x \to c} f(x) = f(c)$.

1.4 The Derivative as a Limit of the Difference Quotient

The instantaneous rate of change (derivative) of $f(x)$ at $x = a$ is defined as the limit of secant line slopes:

```
f'(a) = lim_{h->0} [f(a + h) - f(a)] / h
```

Geometrically, as h approaches 0, the secant line passing through $(a, f(a))$ and $(a+h, f(a+h))$ rotates into the tangent line at $x = a$.